Hongyu Deng

superdhy6@163.com | +1 610 570 1861

EDUCATION

Lehigh University Pennsylvania, United States
P.C. Rossin College of Engineering and Applied Science
Master's in Electrical and Computer Engineering Expected May 2027

Maynooth University Maynooth, Ireland
BEng in Electronic Engineering, Joint Degree Program Sep 2021 - Jun 2025

Fuzhou University Fuzhou, China
Maynooth International Engineering College
BEng in Electronic Information Engineering Sep 2021 - Jun 2025

RESEARCH AND ACADEMIC PROJECTS

Design and Optimization of a Four-Bit Absolute Comparator Mar - Jul 2024

- Designed and integrated absolute-value and comparator logic using truth tables, Karnaugh maps, and De Morgan's laws; verified circuit functionality through MATLAB simulation.
- Adjusted gate sizes and supply voltage along the critical path to evaluate the energy-delay tradeoff.
- Reduced energy consumption by 48.13% at 1.5 times the baseline delay while maintaining circuit functionality.
- Project accepted by the 2024 International Conference on Computers, Information Processing and Advanced Education (CIPAE).

Digital Integrated Circuits and Introduction to VLSI Mar - Jul 2024

Online Research Seminar

- Studied MOS transistor I-V and RC models, inverter voltage transfer characteristics, CMOS scaling, and logical effort for digital logic analysis and gate sizing.
- Completed a two-phase logic design and optimization project, with an oral presentation and written report.

Python Network Monitoring Application Sep 2022 - Jan 2023

- Built a Python application with a tkinter interface for real-time network monitoring, allowing users to specify target IP addresses and ports.
- Integrated iperf and ping for bandwidth and latency measurements; used matplotlib and seaborn to visualize network metrics, including frequency-domain plots.

INTERNSHIP EXPERIENCE

Xiamen Deep Blue Power Technology Co., Ltd. Mar - Jun 2024

Electrical Control Engineer, Technical Department

- Assisted with circuit board soldering, testing, troubleshooting, and repair for charging stations.
- Supported installation and testing of a home backup energy storage system; implemented multithreading and algorithm optimizations to improve performance.
- Contributed to a battery pack module upgrade to address overheating in charging stations; conducted testing and data analysis to support product efficiency and reliability.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB
Tools and methods: tkinter, matplotlib, seaborn, iperf, ping; digital logic design, Karnaugh maps, logical effort, circuit simulation, circuit board testing